# Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Full-Stack · Python · Java · TypeScript · React · Distributed Systems · RAG · LLMs · AI Agents · AWS · PostgreSQL

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## PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years delivering full-stack production systems at enterprise scale - backend APIs, data pipelines, distributed async processing, microservices, and customer-facing features across a multi-tenant SaaS platform serving 100K+ monthly active users. At Entrata (PropTech/Affordable Housing SaaS), owned business-critical systems for 7+ years: the annual HUD compliance data pipeline transforming federal data into ~180,000 domain records, RabbitMQ-based event-driven async processing, AWS S3 document migration with PHI compliance, and 200+ SQL query optimizations across complex multi-tenant PostgreSQL schemas - 99.5% uptime, zero compliance violations, zero security incidents. Shipped 20+ full-stack production modules (Python, Java, PHP, TypeScript, React, PostgreSQL) across 120+ Agile sprints; led microservices redesign achieving 50% scalability improvement; mentored 5+ engineers. In addition, built two production AI systems using RAG pipelines, LLMs, and multi-agent frameworks: **Leasel** (8-agent LangGraph platform with hybrid search, citation-grounded RAG, HITL escalation, and CI regression harness) and **AgentProbe** (open-source LLM evaluation toolkit routing across five providers). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**Languages & Frameworks:** Python (8yr), Java, Spring Boot, TypeScript, Node.js, JavaScript, React, Redux, PHP, Laravel, C#, .NET, Kotlin, Scala, SQL, PL/SQL, Bash, HTML/CSS, GraphQL, FastAPI, Django, Flask

**Databases & Data:** PostgreSQL, MySQL, Oracle, Redis, SQLite, NoSQL, MongoDB, Snowflake, ETL, Data Pipelines, Schema Design, Query Optimization, Stored Procedures, Data Modeling, PySpark, Pandas, NumPy

**Cloud & Infrastructure:** AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, RabbitMQ, Kafka, Celery, GitHub Actions CI/CD, Terraform, Prometheus, Grafana, OpenTelemetry

**AI & ML:** LangGraph, LangChain, RAG Pipelines, Hybrid Search, Embeddings, Vector Databases (ChromaDB, Pinecone, Weaviate), LLM APIs (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama), MCP, Agent Orchestration, HITL, LLM Evaluation, LLM-as-Judge, PyTorch, scikit-learn, BERT, spaCy, HuggingFace, Claude Code

**Practices:** Distributed Systems, Microservices, Multi-Tenant Architecture, REST APIs, OAuth, CI/CD, Agile/Scrum, TDD, Code Review, PHI/HIPAA Compliance, Performance Engineering, Observability

## PROJECTS

### Leasel - Multi-Agent AI Platform with RAG, Memory & HITL

*Multi-Agent · RAG · HITL · Production AI*

Python · LangGraph (8-agent) · FastAPI · TF-IDF + Semantic RAG · OCR · SQLite · CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three full-stack surfaces sharing one SQLite knowledge repository (documents, settings, incidents, release notes, audit logs). **Document Intelligence:** OCR-fallback ingestion, section-aware chunking, hybrid search (TF-IDF + semantic reranking), citation grounding to exact source clauses, missing/conflicting clause detection. **Agentic Investigation:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with durable state, conditional routing, and HITL escalation when confidence is insufficient. **Self-Service:** policy questions answered directly; eligibility and financial decisions always route to staff - enforced in code. CI regression harness with LLM-as-judge scoring; 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

### AgentProbe - Open-Source LLM Evaluation & Observability Toolkit

*LLM Evaluation · Multi-Provider Routing · CI/CD · Open Source*

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions CI/CD · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama | github.com/venkatalakshmiparimala/AgentProbe

15-endpoint FastAPI backend with multi-provider LLM routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama); 5-metric eval engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression harness with golden datasets and LLM-as-judge scoring; offline + online evaluation loops; zero vendor lock-in, runs fully locally. Used to evaluate Leasel across 1,240+ test cases with structured failure analysis and trace-level observability.

### AI on FHIR - HIPAA-Compliant Clinical NLP & API Integration

*Full-Stack · NLP · Enterprise APIs · HIPAA*

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack cloud-native platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60%; HIPAA-compliant RBAC with full data isolation; containerized with Docker. Demonstrates enterprise API integration with AI-powered NLP at compliance-grade reliability.

### CrashClarity - Large-Scale Road Accident Analysis (104K Records)

*Data Engineering · ML · Visualization*

Python · R · scikit-learn · Random Forest · Logistic Regression · GitHub Pages | Portfolio

End-to-end ML pipeline over 104K+ UK road accident records - feature engineering across weather, road type, time, and vehicle variables; Random Forest + Logistic Regression ensemble (85%+ accuracy); interactive visualization dashboard deployed on GitHub Pages.

## EXPERIENCE

### Blue Horse Digital

Sep 2026 - Present

**Software Engineer (Volunteer) - Virginia, USA**

- **Full-stack AI application development** - designing and building Python FastAPI backends and React frontends for ML-powered business intelligence features; integrating LLM APIs (Anthropic Claude) for automated report generation, data summarization, and natural-language querying of structured business data.
- **Data pipeline engineering** - building end-to-end pipelines from raw data ingestion through feature engineering, model training (scikit-learn, XGBoost), evaluation, and API-served predictions; implementing data quality checks, schema validation, and lineage tracking across the pipeline.

- **AI system integration** - connecting ML models and AI components to backend services and downstream reporting layers via REST APIs; applying observability and debugging practices (structured logging, error tracing) to improve reliability of data-driven and AI-powered features in production.
- **Collaboration and delivery** - working across data, backend, and frontend teams to translate business requirements into technical designs; writing API documentation, data specifications, and test coverage for delivered features.

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern - Agentic AI Production Deployment**

- Architected and deployed a production agentic AI system (LangGraph, FastAPI, Python) - multi-step reasoning pipeline with RAG over regulatory documents, tool orchestration, validation guardrails, and conditional routing to export or rejection; sole technical owner from requirements to deployment.
- Deployed an MCP server connecting agentic pipeline outputs to downstream enterprise tooling: defined API contracts, authentication boundaries, failure routing, retry semantics, and observability across the integration.

### George Mason University

Aug 2025 - May 2026

**Graduate Teaching Assistant - College of Engineering and Computing**

- Delivered technical workshops to 600+ students across multiple sections of MIS 303 - Python programming (data structures, OOP, automation scripting) and database design (ER modeling, SQL, MS Access); provided one-on-one debugging support and graded assignments with rubric-aligned feedback.

### Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote) · Multi-tenant platform, 100K+ monthly active users**

- **Full-stack production delivery** - shipped 20+ production modules end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints; owned the full lifecycle from requirements and schema design through backend API, frontend UI, test coverage, deployment, and post-release monitoring; maintained bug count below 10 for 6+ consecutive months.
- **Distributed systems and async processing** - built and operated a RabbitMQ-based event-driven async processing system for gross-rent updates across thousands of tenant records; designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents, annual rent-schedule changes, and multi-year capacity growth; led Docker microservices redesign achieving 50% scalability improvement and 30% deployment time reduction; 99.5% uptime across multi-tenant SaaS production.
- **Data pipelines and compliance engineering** - owned the annual HUD income-limit import pipeline for 7 years, transforming federal source data into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ affordable housing units; designed baseline validation, reconciliation, rollback preparation, and post-release monitoring every cycle; zero compliance violations. Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant PostgreSQL schemas for eligibility, financial, and document workflows.
- **AWS infrastructure and security** - led the Affordable Housing module integration into the company-wide AWS S3 document migration: inventoried all document types, mapped legacy records to migration contracts, validated metadata correctness post-cutover; implemented Libsodium-encrypted PHI pipelines encrypting 100% of resident data (phone numbers, SSNs), zero security incidents across the full migration. Deployed and maintained EC2/RDS production environments; built Python/Bash automation pipelines eliminating 80% of manual operational workload.
- **Technical leadership and mentorship** - mentored 5+ junior engineers on system design, production debugging, and code quality; conducted 300+ peer code reviews; built internal debugging tools reducing engineering triage time by 40%; led cross-functional technical delivery with product, compliance, and operations stakeholders; leveraged Claude Code to accelerate engineering delivery and code review velocity.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (PHP, Python, Java, PostgreSQL) for production SaaS including income-limits import (1.6M HUD/Tax Credit records across 5 years), lease certification documents, and compliance workflows; led AWS S3 cloud migration resolving 90% of legacy storage constraints; redesigned legacy document storage schema migrating hundreds of thousands of records with 6 months post-deployment stabilization.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering** — JNTU Anantapur — 2013 - 2017